Solar Energy

Pupil sheet

Name: _____ **Date:** _____

Safety: Never look at the sun. The box is not waterproof: bring it in if it rains.

1. What I see on the screen

Seven cards: **two measured by sensors, five worked out from them** after the word **Model:**. Tick **M** (measured) or **W** (worked out) — the most important question of the lesson.

The card	M	W	The card	M	W
Illuminance (lx)	☐	☐	**Efficiency** (%)	☐	☐
Air temp. (°C)	☐	☐	**Power, 1 m²** (W)	☐	☐
Irradiance (W/m^2)	☐	☐	**Energy, 1 m²** (Wh)	☐	☐
Panel temp. (°C)	☐	☐			

The screen holds only the last two minutes; the day's curve is in the file from **Export data**.

The station log — written once, at the start:
Place: _____ Orientation: _____ Tilt: _____

2. What I do on my own

First I predict; I do not change my prediction after I measure. Tilted by 60°, the sensor will show about _____ of what it showed facing the source squarely,

because _____

_____

The tilt investigation. $\cos\theta$ is already filled in; you measure and work out the ratio. **At the end return to** 0°**:** if it differs much from the first reading, the sun changed and the series is not sound.

Angle θ	$\cos\theta$	**Light (lx)**	**Ratio** $L(\theta)/L(0)$
0°	1.00		1.00
30°	0.87		
60°	0.50		
75°	0.26		
0° (control)	1.00		

The sky while measuring (circle one): clear broken overcast

Two days, compared. The peak says who shone hardest for a moment. **The area says who produced more energy** — and only that matters to a panel.

	Day 1	Day 2	Which is larger?
Sky (one word)			
Peak light (lx)			
Time of the peak			
Area under the curve			

The area, by the trapezium rule: for each pair of neighbouring readings, I multiply their mean by the time between them; then I add up all the results.

3. The conclusion I write

I predicted that, and the data show that _____

_____

At 60° the measured ratio does, or does not, match 0.50 because _____

_____

The day that produced more energy is the one with the larger area, not the larger peak, because

_____

_____

Exit ticket

Which of the numbers on the screen were measured and which were worked out?

_____